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Translating Physics into Intelligence: Machine Learning for Reliable ESP Condition Monitoring

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Abstract

This research focuses on combining physics-based expert rules with machine learning (ML) to improve the detection of failure-related events in Electric Submersible Pumps (ESPs). The goal is to create a generalized framework for abnormal event detection that addresses the challenge of limited labeled data and sensor data, providing a practical solution for ESP management and maintenance planning.

Physics-based rules are applied to real-time historical ESP sensor data to identify potential failure events throughout the ESP lifecycle. These rules generate failure-related event scores, which serve as training data for a supervised ML model. Instead of directly labeling events as normal or failed, the model learns from these physics-derived scores, helping it detect subtle signals even during normal operations. The training framework is model-agnostic and was benchmarked against widely used semi-supervised learning approaches for comparison.

Physics-based rules showed reasonable accuracy in detecting abnormal events but were often affected by overlapping or noisy signals. Training the ML model on the aggregated rule outputs improved event detection, reduced false positives, and enhanced generalization across different ESP configurations (e.g., sensor availability) and operating environments (e.g., various reservoirs and wells). This approach also increases robustness against sensor faults and adapts to varying operational conditions without extensive retraining.

The final model acts as a denoiser and generalizer of the expert rules, offering reliable early warnings of ESP abnormalities. This makes it suitable for field deployment with minimal manual data labeling or intervention.

This work introduces a hybrid approach where ML enhances and refines physics-based expert systems. By using ML as a denoising layer over the physics rules, the method bridges rule-based and data-driven approaches. It provides a scalable, label-efficient solution for ESP health monitoring, with potential for broader use across the energy sector.

Introduction

ESP reliability remains a decisive lever for production uptime, yet unplanned shut-downs still account for a significant share of lifting costs. Years of field digitalization have turned each pump into a high-frequency data source, but limited ground-truth failure labels and heterogeneous sensor layouts still hinder the move from reactive maintenance to predictive decision-making.

This study tackles that constraint by treating physics knowledge as a source of weak supervision. A library of expert rules—encapsulating known degradation signatures—runs continuously on sliding windows of normalized sensor trends, producing numeric scores that act as soft labels. Rather than stopping at these rule outputs, we train a model-agnostic supervised learner to denoise and generalize the aggregated scores across wells with varying sensor availability and operating envelopes. The resulting model serves as a single, field-deployable anomaly index that requires no manual labeling while retaining the interpretability of its physics roots.

The remainder of the paper sets the technical context, details the rule engine, label-generation pipeline, and learning architecture, and then evaluates operational readiness before outlining future extensions.

State-of-the-Art in ESP Condition Monitoring and Predictive Maintenance

Sensor Landscape and Data Challenges

Modern Electric Submersible Pump (ESP) systems stream a rich mix of down-hole and surface measurements, including intake/discharge pressure, temperature, vibration, motor current, and voltage, over SCADA networks, generating high-frequency multivariate time-series data. The central challenge is transforming this noisy, high-dimensional feed into reliable early warnings of incipient failures ([Ahmed et al. 2019](#)).

Rule-Based and Physics-Driven Foundations

Historically, operators relied on threshold alarms derived from first-principles rules (e.g., motor over-temperature or pressure collapse). These schemes are transparent but prone to false positives when multiple effects overlap. Recent work augments digital twins that update physics simulations in real time, tracking component degradation ([Ahmed et al. 2019](#)). [Rasouli and Pishgar \(2022\)](#) showed that embedding physics-informed constraints inside predictive models preserves physical plausibility, while [Lastra Melo, González, and Anaya \(2023\)](#) coupled wear-and-drag damage models to an LSTM, improving interpretability yet still struggling with cross-field variability.

Supervised Learning Advances

Where labeled failure data exist, diverse supervised algorithms have been deployed. Tree-based ensembles dominate: [Yang et al. \(2022\)](#) reported a 14 % boost in minority-class recall from an RF + XGBoost stack. LightGBM ([Li et al. 2024](#)) and CatBoost ([Molina et al. 2023](#)) now rival classic boosters and provide SHAP-style explainability. Kernel SVMs remains useful for compact feature sets, while deep temporal models such as 1-D CNNs, InceptionTime, and Temporal Fusion Transformers capture long-range dependencies; [Zhao and Chen \(2024\)](#) achieved 48 h look-ahead fault alerts in a Middle East field trial. Emerging techniques such as focal or class-balanced loss and continual-learning updates mitigate severe class imbalance and concept drift, yet periodic retraining is still the norm.

Unsupervised and Semi-Supervised Strategies

Because ESP failures are relatively rare, many studies focus on modeling normal behavior and flagging deviations. Classical methods such as PCA combined with boosting classifiers have been applied to ESP operating data, helping reduce false positives compared to threshold-based alarms ([Abdalla, Salim, and Farouk 2022](#); [Peng et al. 2021](#)). Physics-guided rules have also been integrated with machine-learning

models to preserve interpretability while improving predictive accuracy (Rasouli and Pishgar 2022). Deep autoencoders and LSTM-based variants are widely used for reconstructing multivariate time-series signals and detecting anomalies in industrial systems, including pumps and process equipment (Malhotra et al. 2016). Digital-twin frameworks extend this idea by continuously updating physics simulations with real-time data, enabling tracking of ESP performance degradation and early warning of failure (Ahmed et al. 2019).

Hybrid Physics-Informed Machine Learning

The cutting-edge fuses first-principles insight with data-driven adaptability. In digital twin stacks, real-time sensor feeds refresh physics simulators while an ML layer filters residual noise (Ahmed et al. 2019). Lastra Melo et al. (2023) showed that injecting wear-and-drag indices into an LSTM cut false alarms by 30%. Physics-guided loss terms ensure outputs respect affinity laws (Rasouli and Pishgar 2022). Probabilistic graphical models such as Bayesian Networks encode expert causal maps, for example pressure and current driving efficiency, yielding confidence-scored forecasts. Field pilots and recent research confirm the value of combining physics-based models with machine learning in ESP and industrial monitoring contexts. For example, a general-purpose digital twin framework has been proposed that integrates online learning, transfer learning, Bayesian inference, and Monte Carlo simulation to continuously adapt to changing operating conditions (Ahmed et al. 2019; Zhou et al. 2024).

Methodology

Hernandez et al. (2024) introduced a two-stage hybrid pipeline for ESP failure related abnormal-event handling:

1. Semi-supervised detection — an LSTM auto-encoder trained on expert-labelled "normal" periods flags any reconstruction-error spikes as abnormal events.
2. Physics Rule-based classification — a library of physics-inspired rules diagnoses each flagged event.

That workflow proved accurate yet labor-intensive, because SMEs had to annotate large "normal" spans up front.

Our contribution is to invert the sequence:

- Run physics rules continuously—on every sliding window, not just alarms.
- Treat their numeric scores as weak labels.
- Train a supervised model to denoise and generalize these scores, eliminating the need for manual "normal" tags.

The result is a label-efficient, physics-guided ML framework that delivers earlier, cleaner abnormal failure related event detection.

Motivation for Physics-Based Diagnostic Rules

Early alarm systems successfully highlighted operational changes—such as pump-speed adjustments or choke-setting shifts that affect wellhead pressure—and revealed issues beyond the pump, including gas leaks in tubing, Y-Tool anomalies, and reservoir damage. By detecting these broader abnormalities, they enabled comprehensive monitoring of the entire well system rather than focusing solely on the ESP.

For each predefined alarm category, the rule engine calculates a similarity score that quantifies how closely the current sensor-trend pattern aligns with that category's signature. The category with the highest score is flagged as the primary diagnosis for the window. Ranking alarms in this way surfaces the

most probable root cause while still preserving secondary scores for additional context, thereby reducing ambiguity when different failure mechanisms share overlapping symptoms.

The physics rules sharpen this monitoring in two ways: (1) they cut down needless alarms, and (2) they guide the crew to the right spot. If a rule alarms with a pump-only signature, engineers focus on the ESP. If it alarms with a non-pump signature, they inspect surface equipment, tubing, or the reservoir. In short, the rules broaden ESP management to the entire well while keeping alerts relevant.

Data Preparation – Normalized Change Ratio

All ESP and surface sensors in [Table 1](#) are first resampled to a common interval (user-configurable). For every sensor X we then compute a Normalized Change Ratio (NCR) over a configurable reference window of length W hours (default 12 hours):

$$NCR_t = \frac{x_t - \bar{x}_{t,12h}}{|\bar{x}_{t,12h}| + \epsilon}$$

Where $\bar{x}_{t,12h}$ is the rolling median across the previous 12 hours and $\epsilon = 10^{-6}$ prevents division by zero.

Table 1—List of real-time sensors ESP and surface.

#	Tag name	Origin	#	Tag name	Origin
1	ESP_VIBRATION	Downhole	9	ESP_DISCHARGE_PRESSURE	Downhole
2	ESP_INTAKE_PRESS	Downhole	10	ESP_PASSIVE_LEAKAGE	Electrical
3	WHT	Surface	11	ESP_DIFF_PRESS	Downhole
4	ESP_MOTOR_TEMP	Downhole	12	ESP_CURRENT	Electrical
5	ESP_INTAKE_TEMP	Downhole	13	ESP_FREQUENCY	Electrical
6	WHP	Surface	14	ESP_VOLTAGE	Electrical
7	ESP_ACTIVE_LEAKAGE	Electrical	15	ESP_VOLTAGE_UNBALANCE	Electrical
8	CHP_A	Surface	16	ESP_CURRENT_UNBALANCE	Electrical

We denote this dimensionless value as the Normalized Change Ratio (NCR). Values are clipped to the range $[-1,1]$ to stabilize the downstream rule-membership functions. No other feature scaling is applied; all subsequent steps operate on these normalized trends.

Physics-Rule Engine

The rule library contains 14 membership functions (see sample in [Table 2](#)) corresponding to key failure or operation-change patterns. Each membership function maps one or more NCR inputs to a score in the range $[-1,+1]$:

- $+1 \rightarrow$ strong evidence the window matches the event signature.
- $-1 \rightarrow$ strong evidence the window is inconsistent with that event.

Table 2—Sample score of physics rules for each event.

Event	Score example
YTOOLTUB_FAIL	−0.21
VALVE_LEAK	−0.22
SSSV_CLOSED	−0.14
RESERVOIR_INFLOW_DAMAGED	−0.33
SHAFT_BROKEN	0.07
WORN_PUMP	−0.47
PUMP_STUCK	−0.31
ELECTRICAL_ISSUE	−0.05
FREQ_INCREASE	0.00
FREQ_DECREASE	0.00
WHP_INCREASE	0.19
WHP_DECREASE	0.25
VOLT_INCREASE	0.24
VOLT_DECREASE	−0.08

Rules are sensor-agnostic - if a channel is unavailable for a given well, its contribution is skipped, and remaining inputs are renormalized. Figure 1. shows example membership-function shapes for six representative sensors; the same structure applies across all rules.

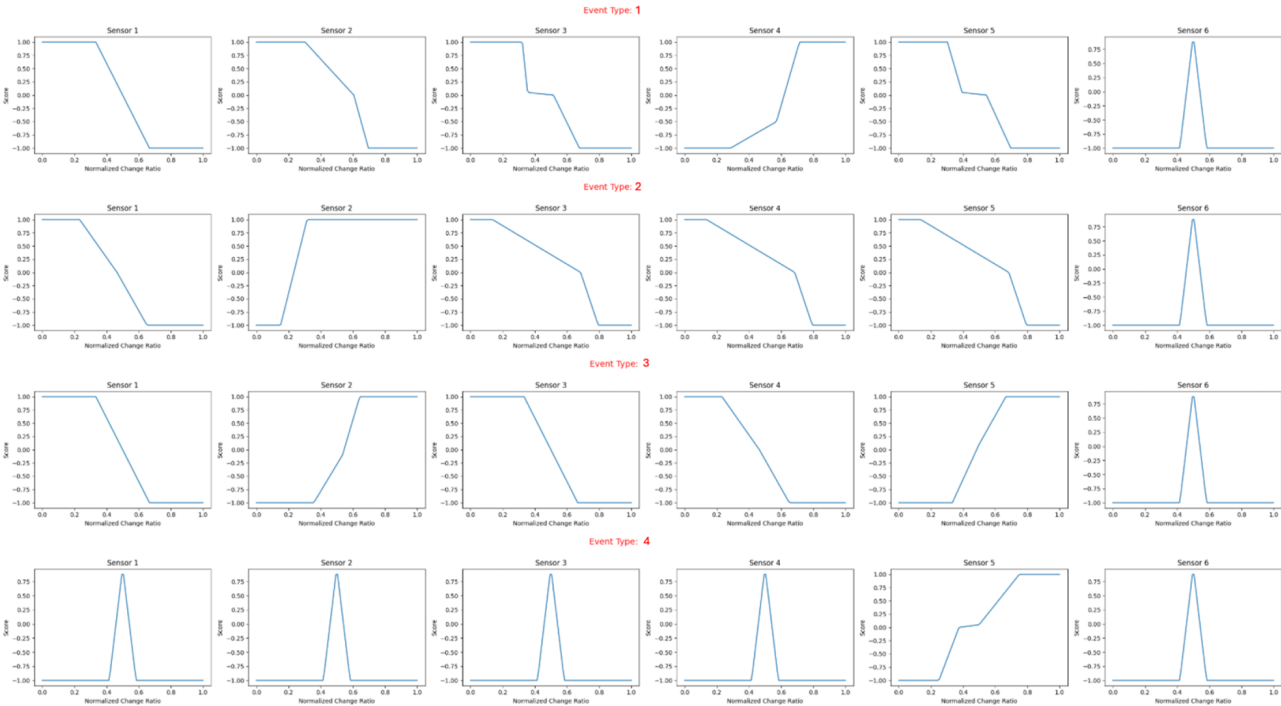


Figure 1—Illustration of physics-rule membership functions and their assigned scores.

Comments: Scores above illustrate one sliding window; real values vary through time.

Physics Scoring & Aggregation & Annotation

The following steps were applied to get the annotations:

1. Windowing – The rules are evaluated on a configurable fixed-length sliding window (window size and stride chosen by the user, e.g., 120 min window / 30 min stride).
2. Activation threshold – Any rule score ≥ 0.10 is considered active evidence of that event.
3. Aggregation – Active scores in the same window are summed to form an aggregated anomaly index A_t . The philosophy: genuine failures tend to excite multiple rules simultaneously.
4. Temporal smoothing – A rolling sum or mean of A_t (e.g., 12-h horizon) yields a stable series S_t that suppresses transient noise while preserving genuine degradation trends (Figure 2).

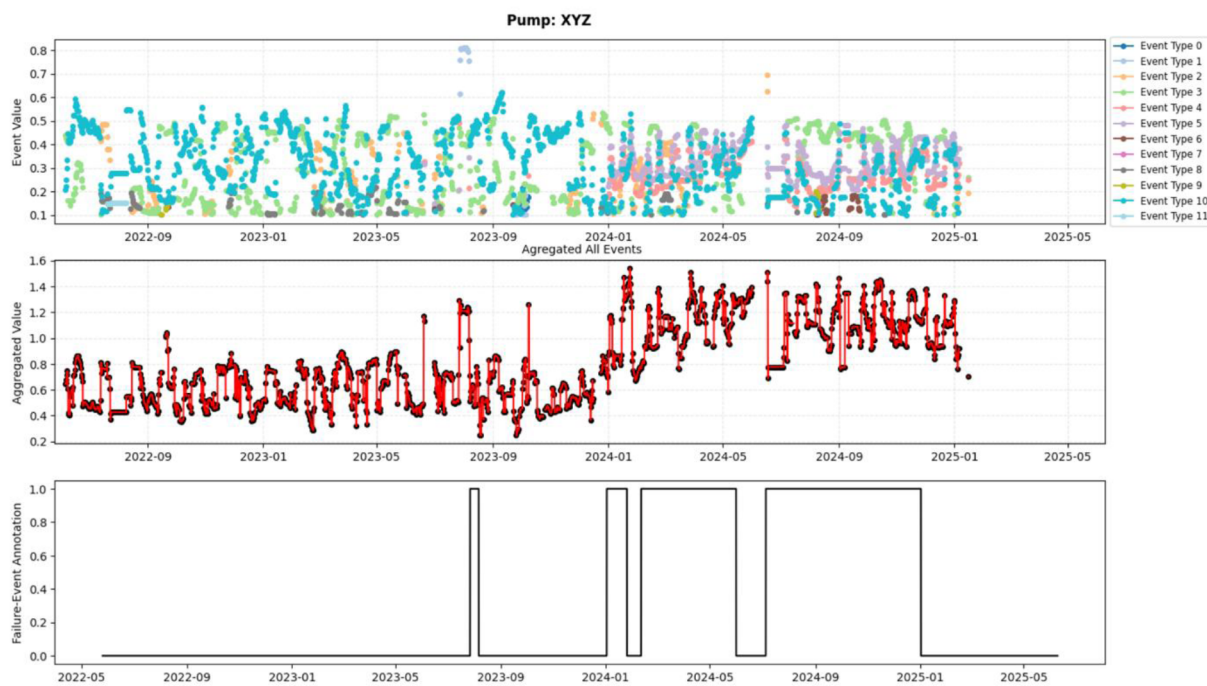


Figure 2—Physics-rule scores, aggregated anomaly index, and resulting failure-event annotations for a single well.

Windows with S_t above a high quantile threshold (e.g., the 95th percentile of the local baseline) are labelled pseudo-abnormal; low-quantile windows become pseudo-normal. Ambiguous middle-range windows are discarded to maintain label purity.

We keep only the time-windows where the physics rules give a high score—periods we are confident show abnormal behavior. These become positive examples; all other windows are treated as normal. A supervised ML model is then trained to tell the two apart. Because its labels come from the physics rules (rather than manual labeling), the model captures expert knowledge while filtering out rule noise. The next chapters will compare three approaches: the raw physics-rule scores, the new supervised-ML scores, and the earlier semi-supervised LSTM-autoencoder results.

Results and Discussion

Evaluation Protocol

Following the evaluation protocol defined by [Hernandez et al \(2024\)](#), we compare three pipelines—(i) the aggregated physics-rule score, (ii) a supervised ML model trained on those rule-based annotations, and (iii) the earlier semi-supervised LSTM-autoencoder.

For every well we compute a weighted F1-score from all its time-window samples; this single value gauges how well each pipeline distinguishes failure-related events from non-failure periods. If a well ultimately experienced a failure, an $F1 \geq 0.70$ is recorded as a correct "failure detected" outcome, whereas an $F1 < 0.70$ is counted as a miss. For wells with no recorded failure, an $F1 \geq 0.70$ is logged as a correct "no-failure" decision, while a lower score indicates a false alarm.

Aggregating these per-well decisions yields a confusion matrix (TP, TN, FP, FN) and an average weighted F1 across all wells. Figure 3 illustrates how time-window labels were prepared before these well-level metrics were calculated Hernandez et al (2024).

Model Evaluation

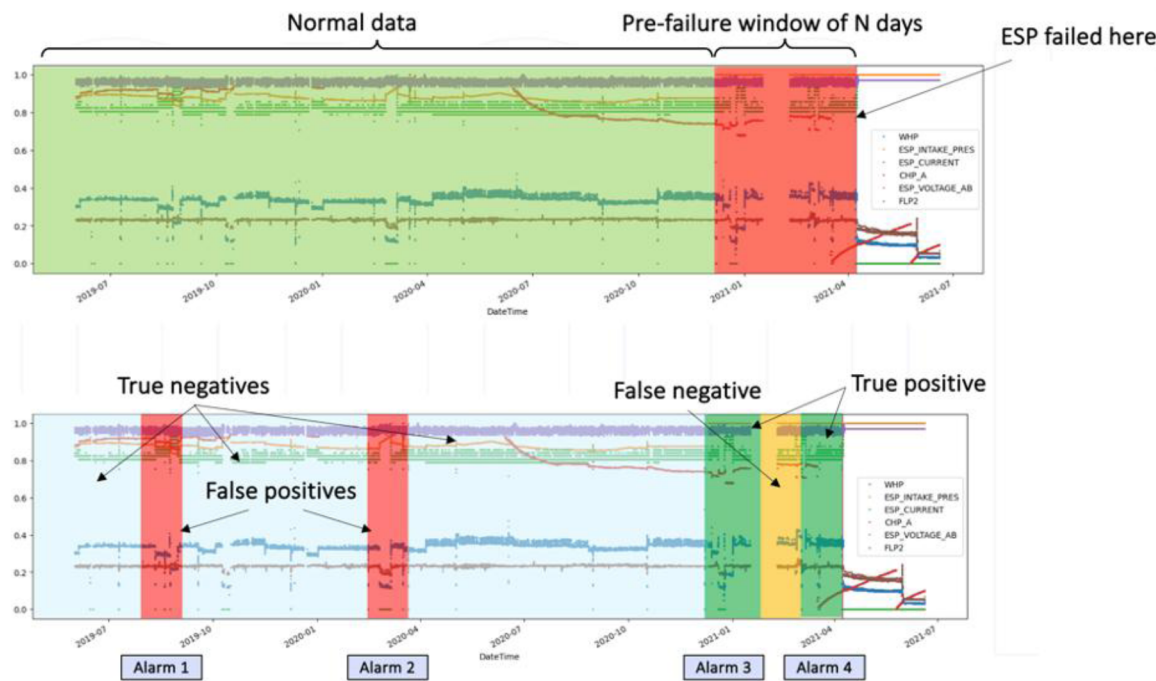


Figure 3—Time-window labels and metrics.

Results

We have used our trained models to inference on 113 pumps real-time historical sensor data. The main objective is to compare methods of labeling on accuracy of the prediction pre-failure periods and failure itself and evaluate performance. In the following 3 confusion matrices of the 3 approaches and accuracy metrics.

Table 3—Confusion Matrix Results - Semi-Supervised Training

Semi-Supervised Training		Prediction Failure Event	
		yes	no
Actual Failure Condition	yes	22	6
	no	4	81

Table 4—Confusion Matrix Results - Aggregated Physics-rule score

Aggregated Physics-rule score		Prediction Failure Event	
		yes	no
Actual Failure Condition	yes	25	3
	no	10	75

Table 5—Confusion Matrix Results - Supervised Training on physics rule-based annotations

Supervised Training on physics rule-based annotations		Prediction Failure Event	
		yes	no
Actual Failure Condition	yes	26	2
	no	7	78

Table 6—Machine Learning Accuracy Metrics

	Accuracy	Precision	TP rate	F1-score (weighted)
Semi-Supervised Training	0.912	0.846	0.786	0.91
Aggregated Physics-rule score	0.885	0.714	0.893	0.88
Supervised Training on physics rule-based annotations	0.920	0.788	0.929	0.918

Aggregated Physics-rule score. Applying the raw, aggregated physics-rule score (middle column) captures the largest share of true failures (TP rate ≈ 0.89) but at the cost of many false alarms, driving precision down to ≈ 0.71 and pulling overall accuracy below 0.89.

Supervised Training: denoising weak labels. Training a supervised learner directly on the rule-based annotations markedly improves accuracy (+3.5 percentage points) and boosts precision to ≈ 0.79 while further increasing recall to ≈ 0.93 . The model effectively "denoises" the weak labels: it learns which rule combinations truly precede failures and which merely reflect benign operating envelopes. This confirms the value of treating physics scores as soft labels rather than hard alarms.

Comparison with semi-supervised baseline. The existing semi-supervised pipeline still edges out the new model on accuracy and precision, but the margins are small (≤ 0.008 absolute). Unlike the semi-supervised approach, the physics-guided pipeline requires no manual curation of normal windows or post-hoc threshold tuning—everything is produced automatically by the rule engine.

Limitations

Abnormal behavior can start as soon as the pump is installed, the example in the [figure 4](#) below (Pump XYZ) illustrates the problem:

- Top panel (Event-type scatter) – Multiple rule types fire almost continuously throughout the pump's life, indicating that several symptoms are present from the very beginning.
- Middle panel (Aggregated score) – The composite anomaly index seldom drops below ~ 0.5 and repeatedly spikes above 1.5, so the pump looks "at risk" for nearly its entire operating period.
- Bottom panel (Failure annotation) – Intermittent failure-event flags appear long before the final shutdown, confirming that the unit operates in a stressed state for months before it ultimately fails.

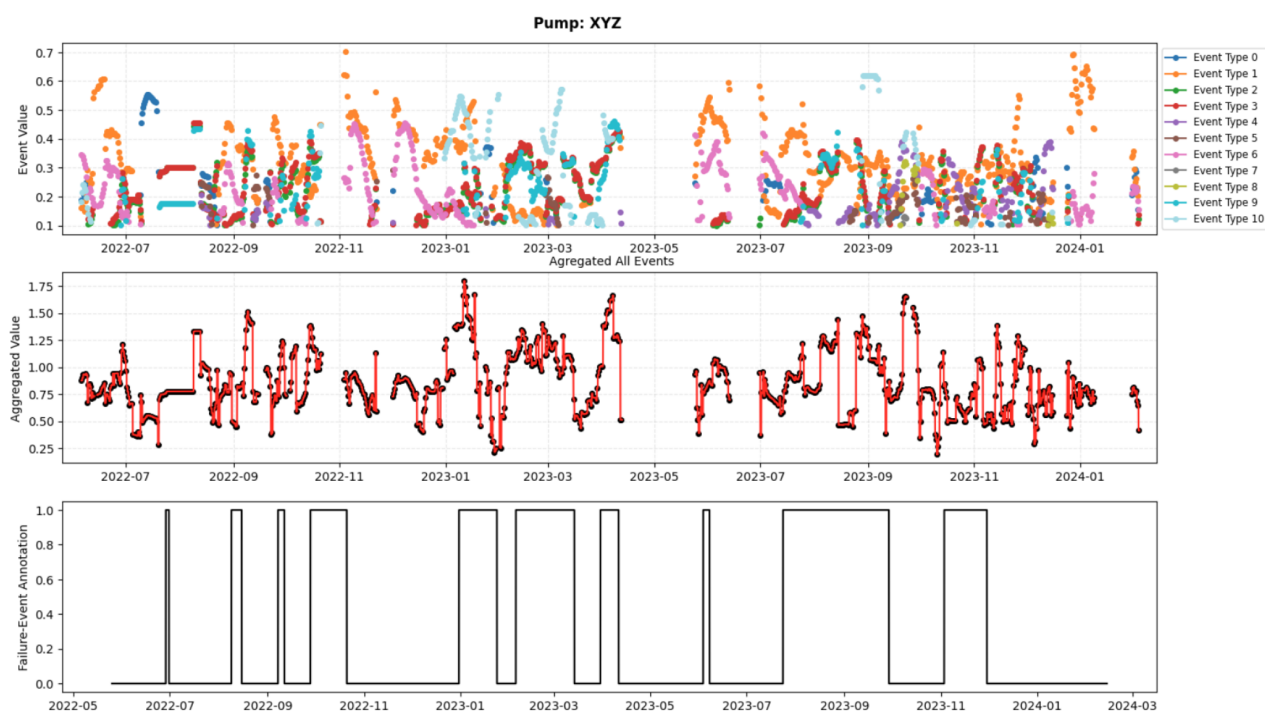


Figure 4—Persistent High Anomaly Scores from Installation to Failure

Persistent high scores can have several root causes introduced during completion or installation—e.g., poor wellbore clean-up, misaligned intake, cable-protector damage. Such wells "struggle" for an extended period, generating frequent alarms that blur the boundary between normal and abnormal behavior. This makes clean labelling difficult and reduces the diagnostic value of simple threshold-based physics rules. Clean labelling therefore becomes difficult, reducing the value of simple threshold rules; consequently, the focus should shift from strict failure prediction toward abnormal-event indication, proactive maintenance, and overall ESP management.

Conclusion

This study demonstrated that treating physics-based rule scores as weak labels and training a supervised learner on top of them can nearly match a fully semi-supervised baseline (accuracy 0.92 vs 0.91) while requiring zero manual labelling of "normal" windows. The supervised model preserved the high recall of raw physics rules (0.93) but cut their false-alarm rate enough to raise precision from 0.71 to 0.79—striking a practical balance between early warning and operational workload.

Field evidence, including wells that show elevated scores from day one, suggests that clean failure boundaries are often ill-defined; therefore, the most valuable output is a reliable abnormal-event indicator that supports proactive maintenance rather than binary failure prediction. By converting continuous physics rule scores into a single, denoised anomaly index, the proposed pipeline delivers exactly that, making large-scale ESP surveillance feasible without increasing human oversight.

Next steps are to expand and fine-tune the rule set, link in installation and well site operation records to explain pumps that always score high, and attach a confidence level to every alarm so crews know how sure the system is; with these upgrades, the physics-guided approach can become a simple, low-cost way to keep ESPs running longer while avoiding needless call-outs.

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